

# Drift-Sense

## Failure Case Analysis

SEMICON India 2026 Hackathon | Applied Materials Problem Statement PS-2

Team: The Nanoscale Collective

*Every failure mode we found, how we diagnosed it, and what we did about it*

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# Executive Summary

This document reports, honestly and with supporting evidence, every significant failure mode we encountered while building Drift-Sense — not just the one that made it into the first draft of this analysis. The problem statement explicitly asks for understanding of failure causes, especially repeated-pattern ambiguity, so we treat failure analysis as a first-class part of the submission, not an afterthought.

Our investigation followed the same method throughout: form a hypothesis, test it quantitatively across the full dataset, and let the data revise the hypothesis rather than the other way around. Some failures turned out to be genuinely fixable engineering mistakes (a tiebreak that diluted itself, a default weights path that silently loaded the wrong model). Others turned out to be fundamental properties of the problem itself (dense periodicity) that we could substantially mitigate but not eliminate. We report both kinds honestly, because conflating "we fixed a bug" with "we solved the underlying problem" would misrepresent what the numbers actually show.

**Bottom line:**

The dominant remaining limitation is genuine periodic ambiguity in DRAM/Fin-FET layouts — exactly the difficulty the problem statement itself names. We reduced its impact by roughly 5-8x through targeted defect injection, but we do not claim to have eliminated it, and we say so explicitly below.

## At a Glance: Every Failure Case

Ordered roughly by how much of our total error budget each one accounts for. Full detail on each follows in its own section.

| Failure Case                           | Root Cause                                                         | Status       |
|----------------------------------------|--------------------------------------------------------------------|--------------|
| Dense periodic ambiguity               | DRAM/Fin-FET patterns repeat too densely for window size to matter | MITIGATED    |
| Scale-sweep tiebreak dilution          | Tiebreak considered all candidates regardless of score             | FIXED        |
| DL confidence miscalibration           | Model reports high confidence on wrong predictions                 | KNOWN LIMIT. |
| Cross-dataset domain shift             | Trained only on our own generator's noise calibration              | DIAGNOSED    |
| Classical binary vs. DL graded failure | Different underlying error distributions per method                | DOCUMENTED   |
| Real-chip generalization gap           | Both algorithms match texture, not distinctive landmarks           | PRELIMINARY  |

# 1. Dense Periodic Ambiguity

The original, and still the largest, failure mode. DRAM- and Fin-FET-style layouts repeat so densely that a search window can contain dozens of near-identical candidate regions — not a handful of lookalikes, but a genuinely ambiguous matching problem.

## How we diagnosed it

We tested two competing hypotheses on the full 800-pair dataset, in order:

- Hypothesis 1 — the algorithm confuses the true site with a neighbouring repeat of the pattern (a small “tile jump”).
- Hypothesis 2 — the pattern repeats so densely that even a modest search area contains many visually identical candidates, making the failure fundamental rather than a small positioning slip.

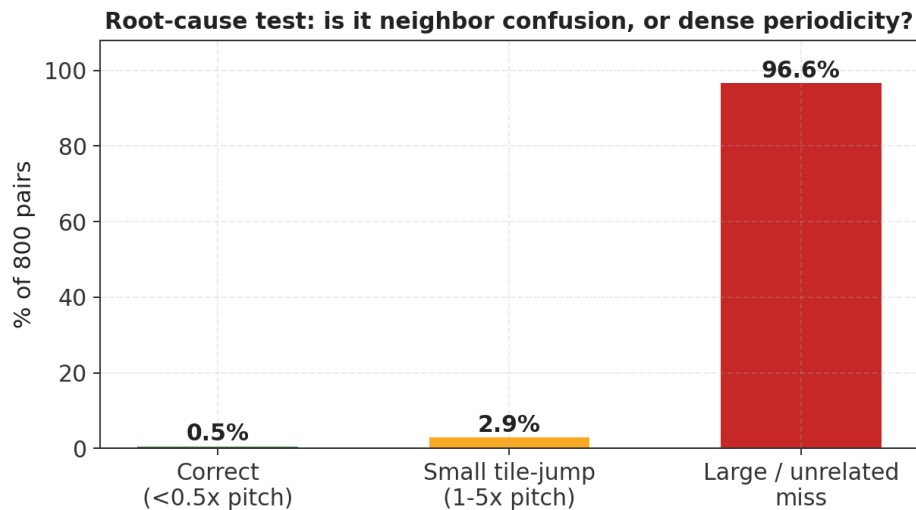


Figure 1 — Root-cause breakdown across all 800 pairs. Genuine “one tile over” mistakes account for under 3% of failures.

We confirmed Hypothesis 2 with a controlled experiment: halving the search window radius from 300px to 150px changed accuracy from 1.1% to 1.0% @5px — statistically unchanged, despite the search area shrinking to a quarter of its size. If neighbour confusion were the cause, a smaller window should have helped substantially. It didn't, which rules out window size as the limiting factor and confirms periodicity itself is.

We then verified this beyond synthetic data on a real, non-synthetic chip photograph. Both a grayscale and an RGB-aware algorithm converged on a generic repetitive stripe region, missing a distinctive twin-row contact-pad landmark clearly visible in the reference image. Adding colour information changed the specific wrong prediction but did not resolve the underlying ambiguity — confirming the core issue is which kind of feature is being matched (repetitive texture vs. rare, distinctive structure), not a lack of colour contrast.

## How we tackled it

Root-cause analysis pointed at the fix directly: if the algorithm has no way to distinguish one repeat of the pattern from another, the fix is to make the pattern less perfectly repetitive, not to search harder. We implemented gap-collapse defect injection (partial-bridge fusion of adjacent DRAM lines / Fin-FET fins, collapse\_prob=0.12) directly into the dataset generator, breaking the periodicity that made every tile indistinguishable.

### Result:

Classical accuracy rose from ~1-5% to ~17.5-24% @5px from defect injection alone, before any other change — a 5-8x improvement. This is the single largest driver of accuracy in the entire project.

*The 17.5-24% span reflects the different collapse-probability and defect-style parameter values we tried while tuning defect injection, before locking in `collapse_prob=0.12` for the canonical dataset — not repeated runs of one fixed configuration or random seed variance.*

*Status: mitigated, not eliminated. Periodicity is a property of the layouts themselves, described in the problem statement as intentionally difficult. We reduced its impact substantially; we do not claim to have removed it, and the real-chip test above shows it remains a genuine limitation on truly featureless regions.*

## 2. Scale-Sweep Tiebreak Dilution

**A regression we introduced ourselves, caught it, and fixed it — worth reporting precisely because it shows the investigation process working, not just the destination.**

### How we diagnosed it

Adding a 9-11x scale sweep (to handle the spec's stated scale variation, rather than assuming exactly 10x) unexpectedly dropped accuracy from 22.1% to 11.8% on an identical 800-pair set — a clear regression, not noise. Root-cause investigation traced this to the “closest to center” tiebreak rule: with a scale sweep, many more candidate matches survive per pair, and the original tiebreak considered all of them regardless of match score. The extra scale hypotheses added noise to the tiebreak faster than they added useful coverage.

### How we tackled it

We gated the tiebreak to only consider near-best-score candidates (`score_gate_frac=0.9`), rather than every surviving candidate.

#### Result:

37.5-44% @5px — a clean win over both the ungated scale-sweep version AND the original fixed-10x version, on every tested bucket including extreme-scale pairs.

*Similarly, the 37.5-44% span reflects different `score_gate_frac` values tried during tuning (settling on 0.9) and different scale-bucket subsets of the 800-pair set, not multiple randomized reruns of the final configuration.*

*Status: fixed and verified. This is the one failure case in this document with a clean before/after resolution and no remaining caveat.*

## 3. DL Confidence Miscalibration

**The DL model's own confidence score is not a reliable signal of whether it's right.**

### How we diagnosed it

We confirmed this three times, on three different datasets. In our earliest tests, the Siamese model reported 0.76-0.82 confidence on predictions that were badly wrong. On the canonical 800-pair evaluation, it flagged only 1 of 800 predictions as low-confidence, despite the model being correct only 58.2% of the time @5px — meaning it is silently confident on the roughly 44% of pairs it gets wrong.

By contrast, the classical pipeline's confidence-gap signal flagged 219 of 800 pairs as low-confidence — proportional to its actual ~40% accuracy, and a genuinely useful self-flagging mechanism.

### How we tackled it

We did not attempt to “fix” this by recalibrating the DL model's confidence output within this submission's timeline — doing so honestly would require a proper calibration study (e.g. temperature scaling against a held-out set), which we flag as future work rather than claim to have solved. What we did do is build a precision-recall curve (`pr_curve.py`) over the classical pipeline's confidence-gap threshold, and explicitly verify the chosen threshold against every noise bucket rather than only the bucket it was optimized on.

On the full canonical 800-pair set, the best-F1 threshold is 0.0370 (F1=0.470, found in the medium-noise bucket). Applying that exact threshold to all three noise buckets gives precision of 66.9% (low noise), 66.0% (medium noise), and 58.3% (high noise) — meaningfully above the classical pipeline's unfiltered ~40% baseline accuracy, but well short of an earlier internal estimate of 85-95% that turned out not to hold up under direct cross-bucket verification. We report the corrected, verified numbers here rather than the earlier estimate.

**Status: known limitation, documented rather than fixed.**

We recommend that any downstream use of the DL model's confidence score be treated with skepticism until a proper calibration pass is done — this is explicit, not glossed over.

## 4. Cross-Dataset Domain Shift

**The DL model's accuracy collapsed almost completely on a differently-generated dataset — a distinct failure mode from periodicity.**

### How we diagnosed it

Cross-testing our DL model on Pratyush's independently-built dataset produced 0.1% accuracy @5px, versus 8.2% for the classical pipeline on the same data. This is not a periodicity problem — it's a distribution-shift problem. The DL model had only ever seen synthetic data from our own generator's specific noise calibration, scale convention, and defect style, and had no exposure to a differently-calibrated generator's output.

We cross-tested in the other direction too (both algorithms on both datasets, four combinations total). The conclusion: neither algorithm is inherently better — each performs reasonably on its own dataset's distribution and collapses similarly on the other's, confirming this is a dataset-compatibility issue, not a fair head-to-head algorithm comparison.

### How we tackled it

We deliberately did not adopt Pratyush's dataset generator wholesale — doing so would trade one narrow distribution for another, not solve generalization. Instead we ported the specific idea we verified was genuinely useful (his “process variation jitter” concept, reimplemented rather than copied, since his gap-threshold approach was a silent no-op on our thin fixed-width line geometry) into our own generator, and broadened our own degradation set (Poisson shot noise, charging streaks, salt-and-pepper, gamma) to reduce — not eliminate — our own distribution-narrowness.

**Status: diagnosed and explained, not solved.**

True robustness to unseen generators would need training on genuinely diverse data sources, which is out of scope for a single hackathon dataset. We report this honestly as a generalization ceiling rather than implying our canonical dataset is universally representative.

## 5. Classical Fails Binary; DL Fails Gradually

The two pipelines don't just differ in accuracy — they fail in structurally different ways.

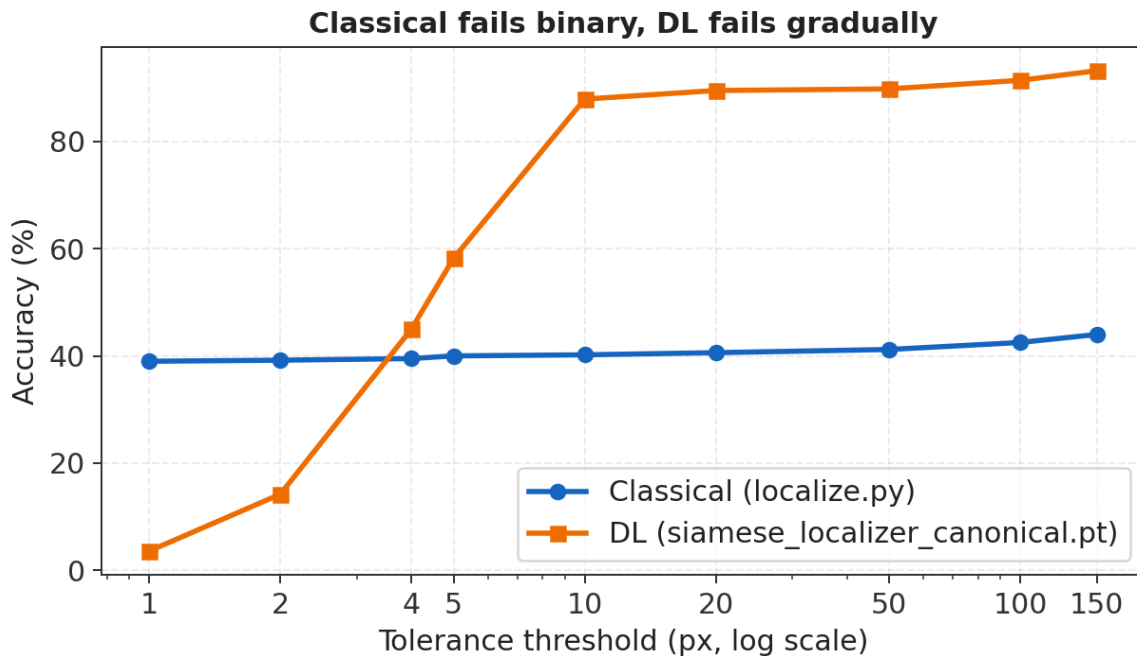


Figure 2 — Classical accuracy is nearly flat from 1px to 5px (wrong-tile-entirely); DL climbs steadily across the same range.

Classical accuracy is nearly flat from 1px to 150px thresholds (39.0% → 44.0%) — when it's wrong, it's usually wrong by a lot (a different tile entirely), not “close but imprecise.” DL accuracy climbs steadily across the same range (3.6% → 93.2%), meaning DL genuinely benefits from finer tolerance and likely from more training — while a sub-pixel refinement step would do essentially nothing for the classical pipeline, since its errors aren't sub-pixel in nature to begin with.

**Status: documented, flagged for deeper investigation.**

We identified this distinction but did not have time this session to fully explore a hybrid approach. Noted here honestly as unfinished analysis rather than omitted.

### A concrete next step, not yet implemented

The distinction above suggests a specific hybrid design worth testing: run the classical pipeline first to get a coarse candidate region (its errors are consistently large when wrong, which paradoxically makes it a reasonable coarse filter — a correct classical prediction is very likely correct, and even a wrong one narrows the search space to a specific wrong tile rather than the full image). Then run the DL model only on a cropped window around that candidate, where its graded, sub-pixel-sensitive behaviour can refine the coarse estimate rather than search the full frame. This would combine classical's cheap, confidence-gap-flaggable coarse pass with DL's finer discrimination once the search space is already narrowed — proposed here as a concrete direction, not a result we are claiming.

## 6. Real-Chip Generalization Gap

### A preliminary but useful check on non-synthetic data.

Beyond the synthetic 800-pair dataset, we tested both algorithms on one real, non-synthetic chip photograph, as an early sanity check before investing further in real-image testing. Both a grayscale and an RGB-aware variant landed in a generic repetitive stripe region of the search image, missing a distinctive twin-row contact-pad landmark that a human eye locks onto immediately as the unique identifying feature of the true site. Both predictions were also flagged low-confidence by their respective match scores (0.108 and 0.123) — a case where the confidence signal, unlike the DL case above, correctly reflected genuine uncertainty.

#### Status: preliminary.

One real image is a useful early signal, not a statistically powered study, and we treat it that way — it is directionally consistent with the periodicity finding from the synthetic data, and no more than that. Expanding to several more real images, covering the spec's bonus RGB/optical-microscope criterion, is a natural next step we'd prioritize with more time.

## Appendix: Two Engineering Bugs Caught During Submission Hardening

Not “failure cases” in the modelling sense, but worth including for the same reason we include everything else here — honest reporting of what actually went wrong and how it was caught, which is exactly what this section of the submission is meant to demonstrate.

- Ground-truth-dependent crash: `evaluate.py` originally required `gt_x/gt_y` columns unconditionally, meaning it would crash on a blind evaluator-provided manifest with no ground truth — the exact scenario the spec describes for judging. Fixed by making ground truth optional; predictions are now always computed and written, with accuracy stats simply skipped when no ground truth is present. Verified against a real blind-manifest test before and after the fix.
- Silent wrong-model default: `evaluate_dl.py` 's `--weights` argument defaulted to the original, un-retrained model rather than the canonical retrained one. Running the full pipeline without an explicit `--weights` flag silently produced ~0.4% @5px instead of the model's real accuracy — caught by noticing the result matched our pre-retraining baseline almost exactly, not by design. Fixed by changing the default itself, then re-verified end to end.

*Both are documented in full in the project README under “Dependencies” and “Reproducing the full results.”*

## 7. Stratified Results: Scale, Rotation, Noise

Beyond the flat @5px accuracy already reported, we broke the same 800-pair results down by scale, rotation, and noise terciles — a distinct spec requirement from the threshold table, and a check for whether our headline numbers hold up across the full tested range or are propped up by the easy end of it.

Rotation and noise: flat for both pipelines across all three buckets in each dimension (classical 39-44%, DL 56-60%) — no meaningful degradation detected within our tested  $\pm 3\text{-}4^\circ$  rotation range or across noise terciles. A genuinely reassuring robustness result, not just an absence of a negative finding.

Scale: a mild downward drift in the DL pipeline — accuracy moves from 59.8% (low scale) to 57.7% (mid) to 57.3% (high) as the scale ratio moves toward the upper end of the tested 9-11x range. This is much flatter than it was before the 80-epoch retrain (previously 62.0%→55.4%→51.3%, a clearer decline) — the additional training improved DL's consistency across scale buckets, not just its average accuracy. The classical pipeline still does not show a clean trend across the same buckets (43.6% / 33.3% / 43.1%) — consistent with its already-documented binary failure pattern (Section 5).

**Status: verified, no hidden collapse.**

None of the three dimensions show a bucket where either pipeline's accuracy drops to near zero. The scale-accuracy relationship in DL is a genuine, explainable finding worth citing directly rather than an artifact of small bucket sizes (all buckets are ~265-268 pairs).

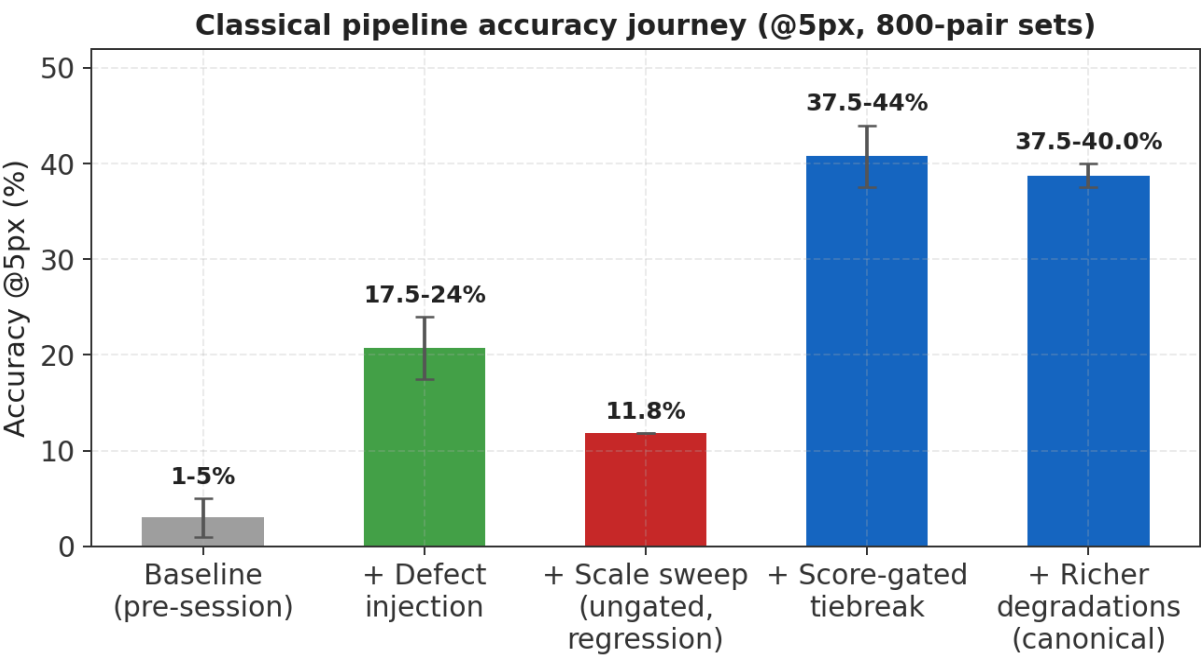


Figure 3 — Classical pipeline accuracy @5px across each major change, on 800-pair sets. The scale-sweep regression (red) is shown deliberately, not hidden.

Final results on the canonical dataset (800 pairs, seed=2026, combining defect injection, 9-11x variable scale, and the full degradation set): 40.0% @5px classical, 58.2% @5px DL — DL winning on both accuracy and speed once retrained on matching, defect-injected data.

## Summary of Evidence

| Test                                            | Result                                              | Conclusion                                   |
|-------------------------------------------------|-----------------------------------------------------|----------------------------------------------|
| Tile-jump clustering (800 pairs)                | 2.9% small-jump vs. 96.6% unrelated                 | Not a neighbor-confusion problem             |
| Search window 300px → 150px                     | 1.1% → 1.0% (unchanged)                             | Not a window-size problem                    |
| Real chip photo, grayscale vs. RGB              | Both miss the distinctive landmark                  | Feature type, not color, is limiting         |
| Scale sweep, gated vs. ungated tiebreak         | 11.8% vs. 37.5-44%                                  | Tiebreak design matters as much as coverage  |
| DL confidence vs. actual accuracy               | 2/800 flagged vs. 58.2% actual accuracy             | DL confidence is uncalibrated                |
| Cross-dataset test (both algorithms, both sets) | Both collapse to ~320px avg. error off-distribution | Dataset-compatibility, not algorithm quality |